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SAP : 56892.

Course : Data Visualization (BS-DS-6-1).

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LAB TASK 02

Data Visualization in Excel

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1. Introduction

This lab focuses on reporting and visualizing insights from the Titanic dataset using Microsoft Excel. Different Excel features such as sorting, filtering, conditional formatting, charts, and pivot tables are used to analyze passenger survival patterns.

2. LAB OBJECTIVES

The objectives of this lab are:

- To explore a real-world dataset
- To perform sorting and filtering
- To apply conditional formatting
- To create different charts
- To analyze data using pivot tables
- To interpret visual insights

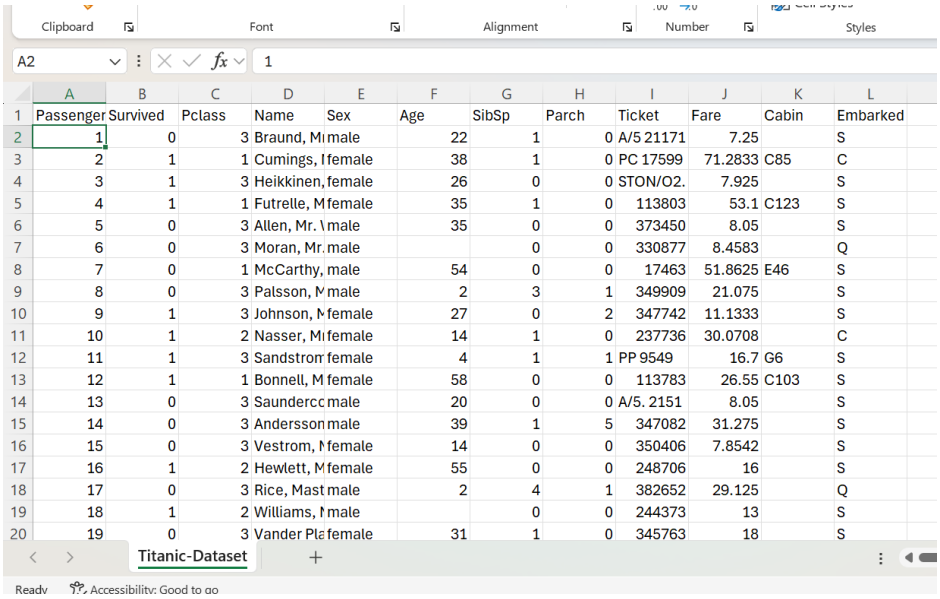
3. DATASET DESCRIPTION

The Titanic dataset contains information about passengers onboard the Titanic.

Key Attributes:

- PassengerId
- Survived (0 = No, 1 = Yes)
- Pclass (1st, 2nd, 3rd)
- Name
- Sex
- Age
- Fare

- Embarked



Passenger	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, M	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, I	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen,	female	26	0	0	STON/O2.	7.925		S
4	1	1	Futrelle, M	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. \	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr.	male		0	0	330877	8.4583		Q
7	0	1	McCarthy,	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, M	male	2	3	1	349909	21.075		S
9	1	3	Johnson, M	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mi	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, M	female	58	0	0	113783	26.55	C103	S
13	0	3	Saunderco	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, M	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, M	female	55	0	0	248706	16		S
17	0	3	Rice, Mast	male	2	4	1	382652	29.125		Q
18	1	2	Williams, M	male		0	0	244373	13		S
19	0	3	Vander Pla	female	31	1	0	345763	18		S

4. TOOLS USED

- Microsoft Excel
- Titanic Dataset (Excel format)
- Microsoft Word (for report)

5. All Tasks

TASK 1: Sorting Data by Age:

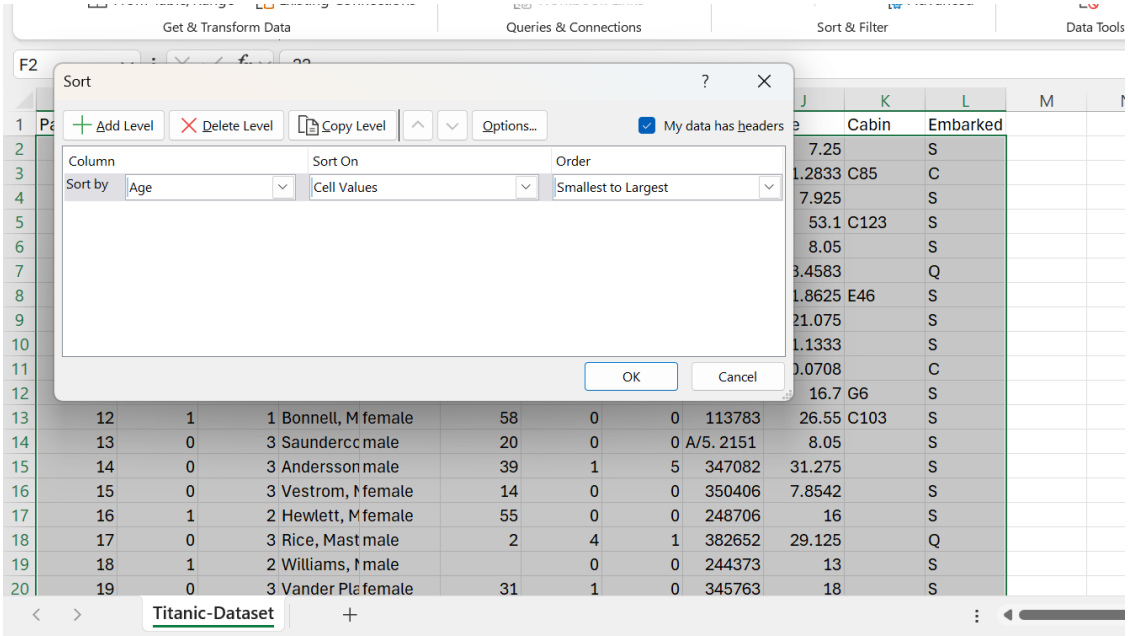
Goal

To arrange passengers from youngest to oldest.

Working (What is happening)

Sorting arranges the data in a specific order. When we sort the **Age column from smallest to largest**, Excel reorganizes all rows so that the youngest passengers appear first and the oldest appear last. All related data (name, class, fare, etc.) moves together correctly.

Screenshots :



C	D	E	F	G	H	I	J	K	
lass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Em
3	Thomas, M	male	0	0	1	2625	8.5167		C
2	Hamalaine	male	1	1	1	250649	14.5		S
3	Baclini, Mi	female	1	2	1	2666	19.2583		C
3	Baclini, Mi	female	1	2	1	2666	19.2583		C
2	Caldwell, M	male	1	0	2	248738	29		S
2	Richards, I	male	1	1	1	29106	18.75		S
1	Allison, Ma	male	1	1	2	113781	151.55	C22 C26	S
3	Panula, Ma	male	1	4	1	3101295	39.6875		S
3	Johnson, M	female	1	1	1	347742	11.1333		S
2	Becker, Ma	male	1	2	1	230136	39	F4	S
3	Nakid, Mis	female	1	0	2	2653	15.7417		C
3	Goodwin, M	male	1	5	2	CA 2144	46.9		S
3	Dean, Mas	male	1	1	2	C.A. 2315	20.575		S
2	Mallet, Ma	male	1	0	2	S.C./PARIS	37.0042		C
3	Palsson, M	male	2	3	1	349909	21.075		S
3	Rice, Mast	male	2	4	1	382652	29.125		Q
3	Andersson	female	2	4	2	347082	31.275		S
3	Strom, Mis	female	2	0	1	347054	10.4625	G6	S
1	Allison, Mi	female	2	1	2	113781	151.55	C22 C26	S

Insights (What we learn)

- We can clearly see the **youngest and oldest passengers.**
- Age-based patterns become easier to analyze.
- It helps in further analysis such as identifying **children and elderly passengers.**

TASK 2: Filtering Survivors:

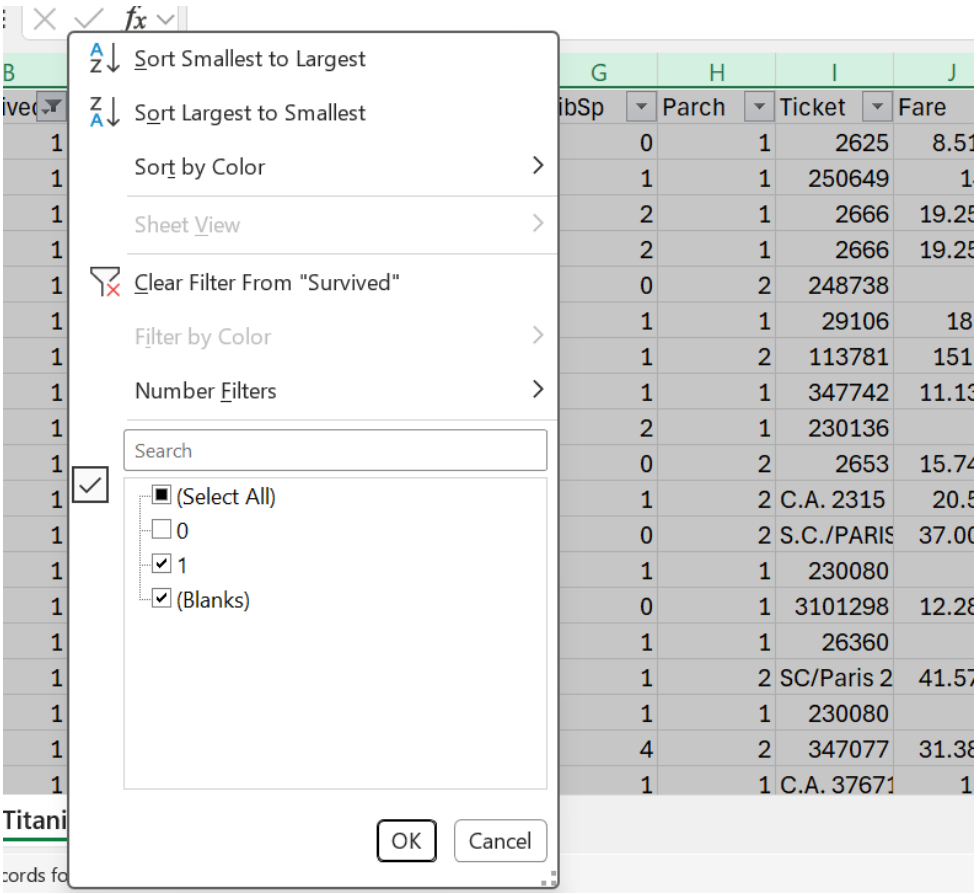
Goal

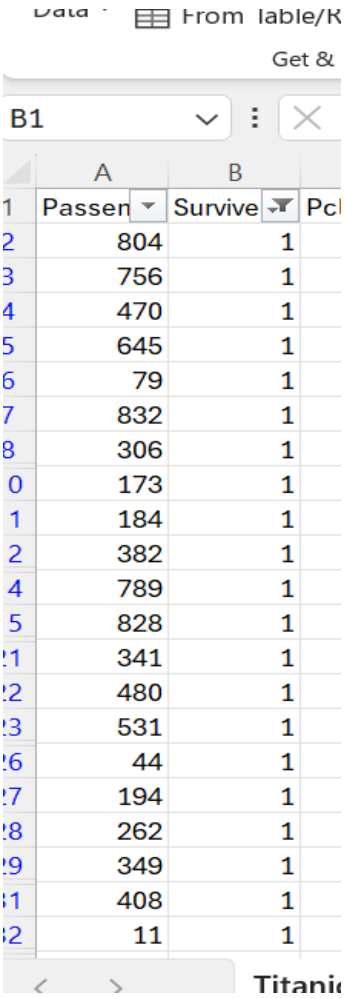
Display only passengers who survived.

Working

Filtering temporarily hides rows that do not meet a condition. When we apply a filter on the **Survived column** and select only value 1, Excel displays only passengers who survived the Titanic disaster while hiding non-survivors.

Implementation (Screenshots)





	A	B	
1	Passen	Survive	Pcl
2	804	1	
3	756	1	
4	470	1	
5	645	1	
6	79	1	
7	832	1	
8	306	1	
0	173	1	
1	184	1	
2	382	1	
4	789	1	
5	828	1	
1	341	1	
2	480	1	
3	531	1	
6	44	1	
7	194	1	
8	262	1	
9	349	1	
1	408	1	
2	11	1	

Insights

- Allows focused analysis on survivors only.
- Helps study characteristics (age, gender, class) of people who survived.
- Makes comparisons easier in later tasks.

TASK 3: Conditional Formatting (Children Highlight)

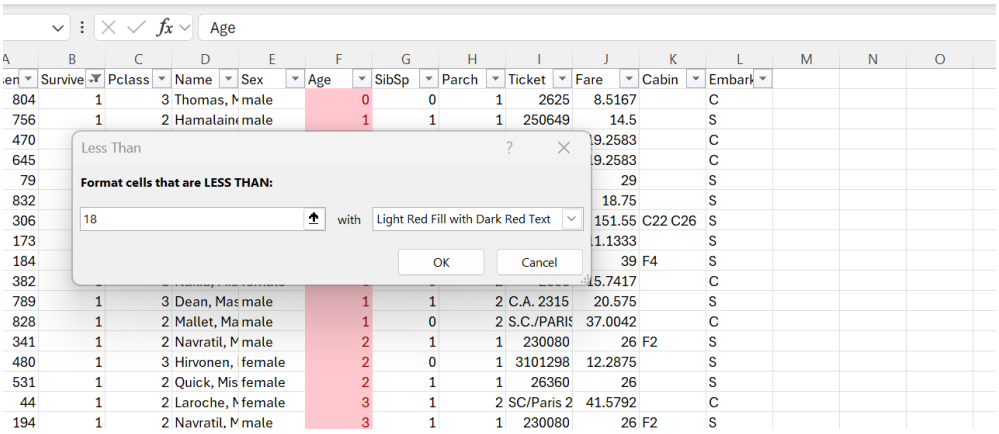
Goal

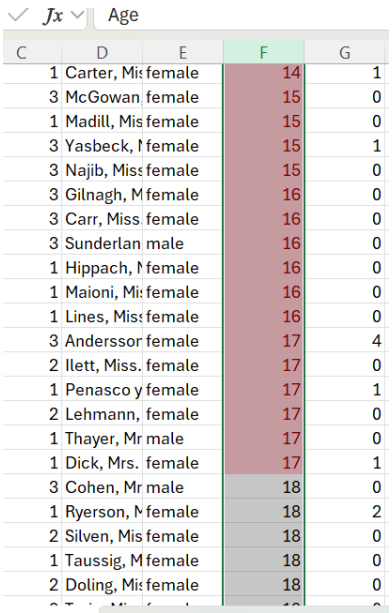
Highlight passengers under 18 years.

Working

Conditional formatting automatically applies color to cells based on a rule. When we apply the rule **Age < 18**, Excel highlights all passengers who were children. The data itself does not change—only the visual appearance does.

Implementation





Insights

- Children are instantly visible in the dataset.

- Helps identify whether children had better survival chances.
- Improves readability and quick decision-making.

TASK 4: Bar Chart – Survivors vs Non-Survivors

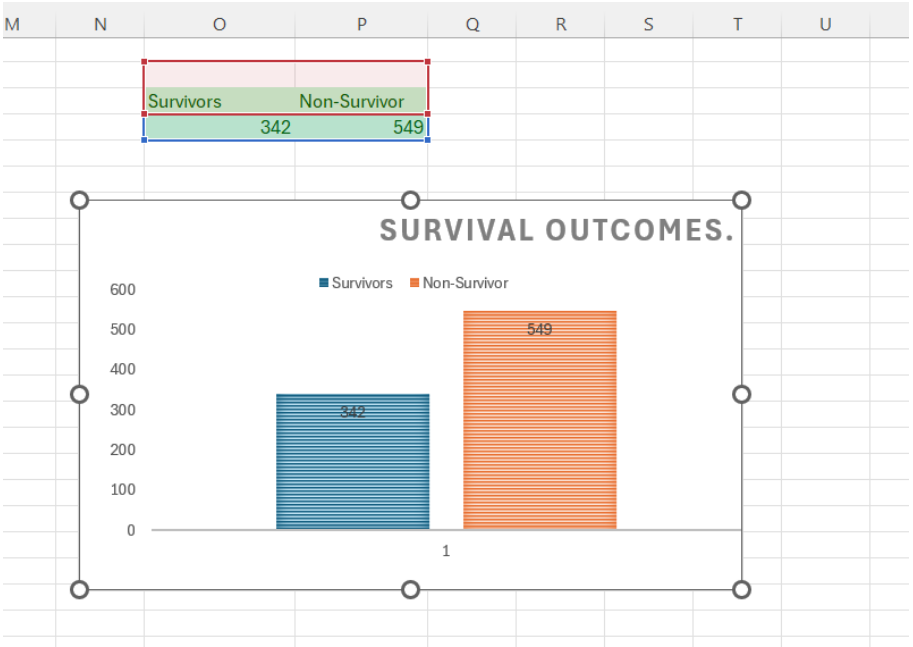
Goal

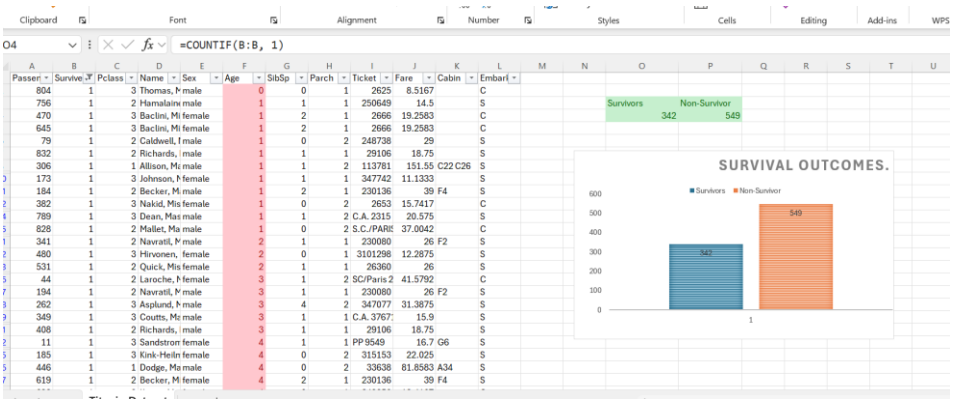
Compare number of survivors and non-survivors.

Working

A bar chart compares numerical values visually. Using the **COUNTIF** function, we count survivors and non-survivors, then represent these counts using vertical bars. The height of each bar represents the number of passengers.

Implementation





Insights

- Non-survivors are significantly more than survivors.
- Shows the severity of the Titanic disaster.
- Visual comparison is clearer than raw numbers.

TASK 5: Pie Chart – Passenger Class Distribution

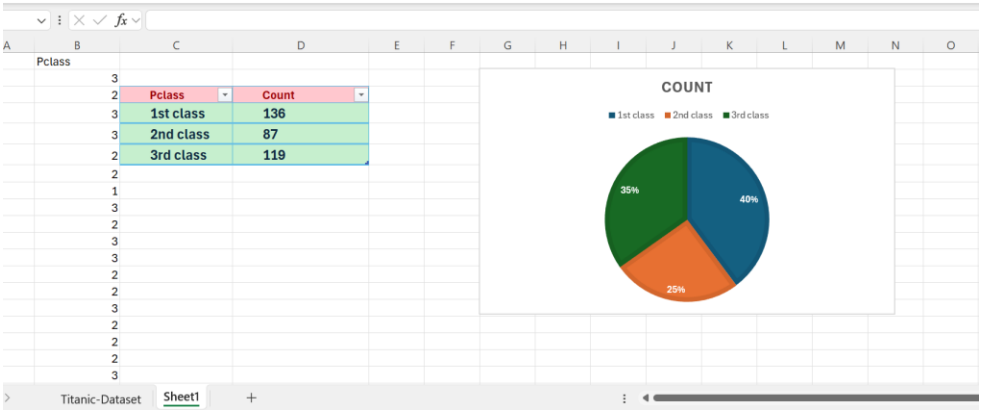
Goal

Visualize passenger distribution across classes.

Working

A pie chart shows how a whole is divided into parts. We count passengers in each class (1st, 2nd, 3rd) and display their percentage share using a pie chart.

Implementation



Insight

- Majority of passengers were from **3rd class.**
- 1st class had the smallest number of passengers.
- Highlights social and economic distribution onboard.

TASK 6: Histogram – Age Distribution

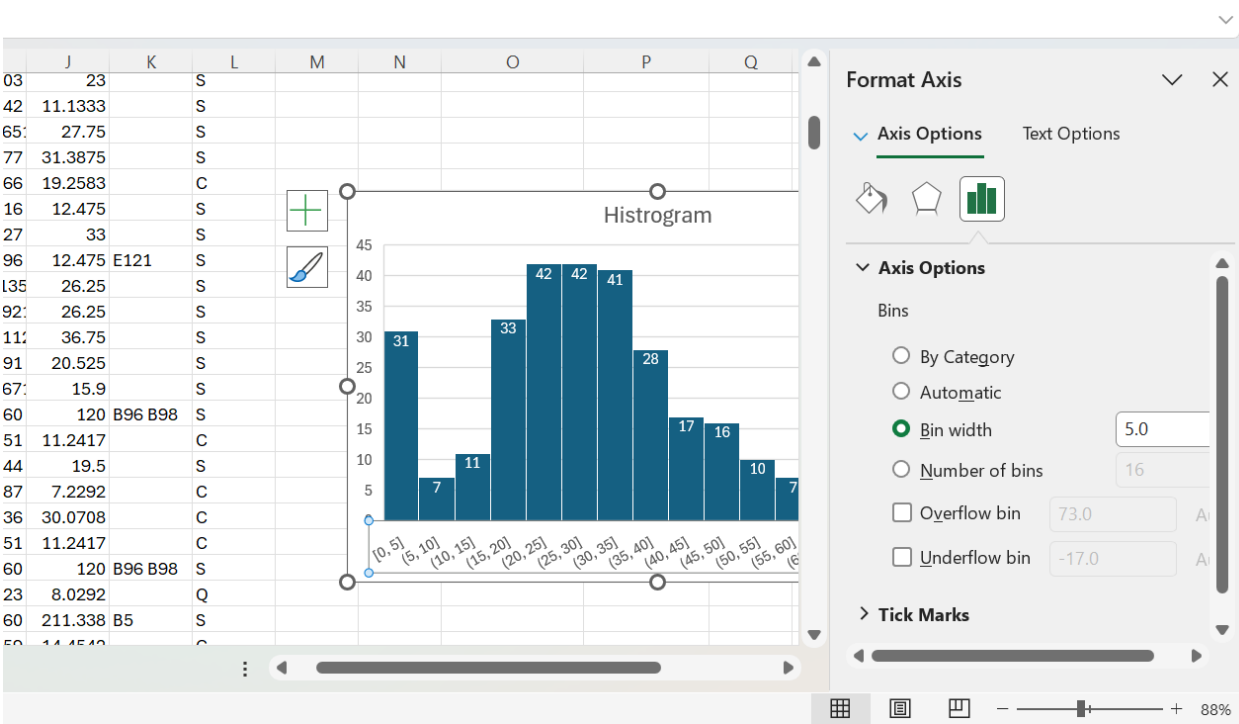
Goal

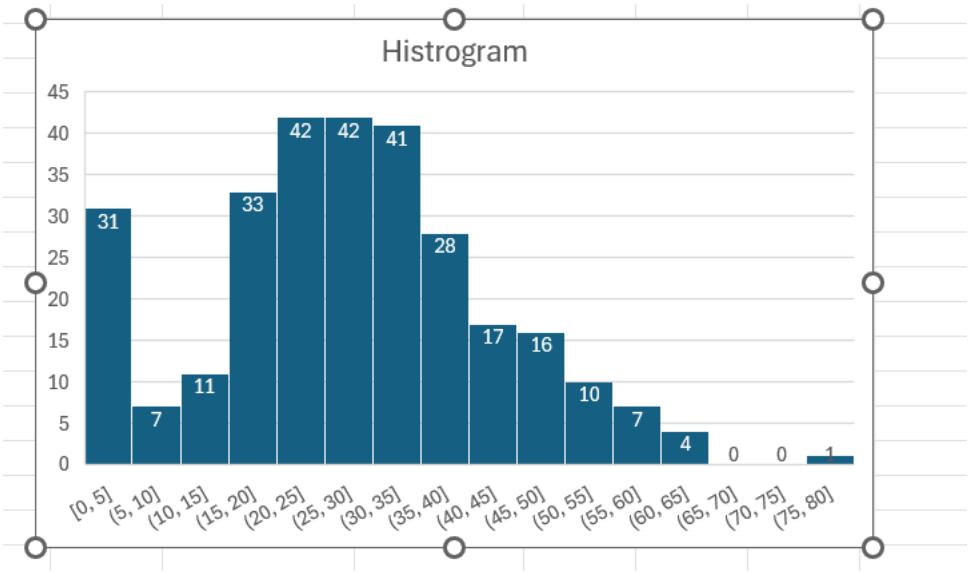
Understand age frequency.

Working

A histogram groups continuous data into ranges (bins). When we create a histogram of age with a bin width of 5, Excel groups passengers into age intervals (e.g., 0–5, 6–10, etc.).

Implementation





Insights

- Most passengers were between **20–40 years.**
- Very few passengers were extremely young or very old.
- Helps understand overall age structure of passengers.

TASK 7: Clustered Bar Chart – Gender Survival

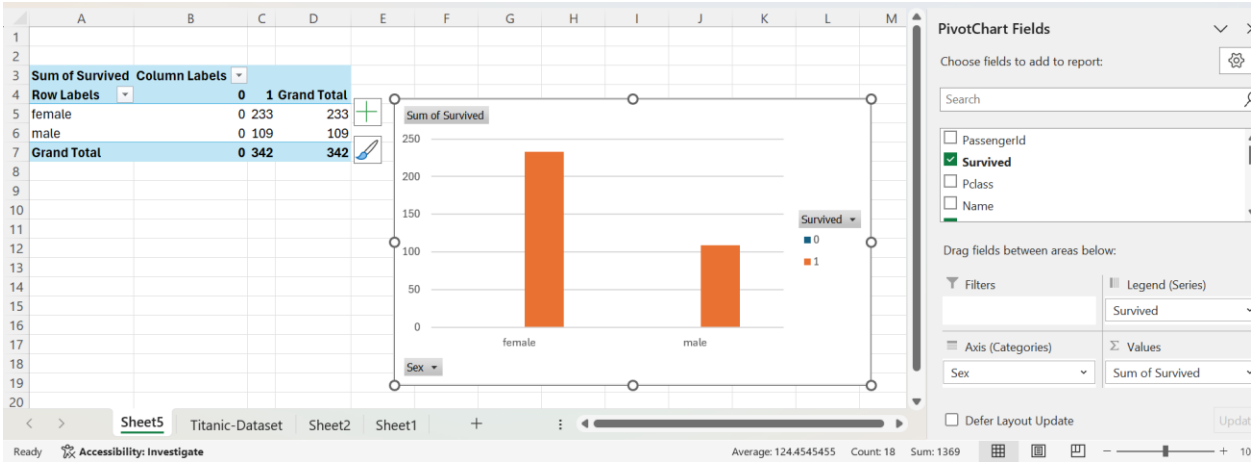
Goal

Compare survival between males and females.

Working

A pivot table groups data automatically. By placing **Sex in rows** and **Survived in columns**, Excel calculates survival counts by gender. The clustered bar chart visually compares survival for males and females.

Implementation



Insight

- Females had a much **higher survival rate.**
- Confirms "women first" evacuation policy.
- Gender played a major role in survival chances.

TASK 8: Line Chart – Age Groups

Goal

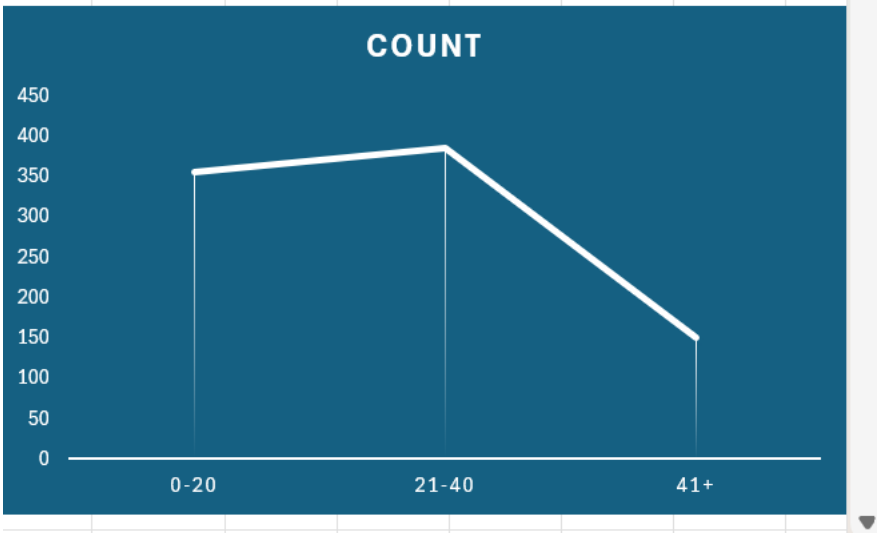
Analyze survival trends by age group.

Working

Age groups are created using an IF formula. Passenger counts in each age group are plotted using a line chart. The line shows changes across age groups.

Implementation

Age Grou	Count
0-20	356
21-40	385
41+	150



Insight

- Younger age groups show higher survival trends.
- Survival decreases with increasing age.
- Line chart helps identify trends rather than just comparison.

TASK 9: Pivot Table – Average Fare by Class

Goal

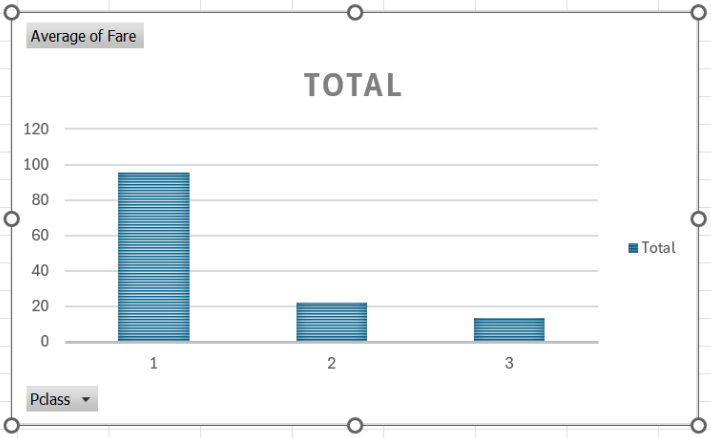
Calculate average fare per passenger class.

Working

A pivot table summarizes large data efficiently. By placing **Pclass in rows** and **Fare in values (Average)**, Excel calculates the average fare paid by passengers in each class.

Implementation

Row Labels	Average of Fare
1	95.60802868
2	22.0557
3	13.69488739
Grand Total	**48.3954076**



Insight

- 1st class passengers paid the highest fares.
- 3rd class passengers paid the lowest fares.
- Shows economic differences among passenger classes.

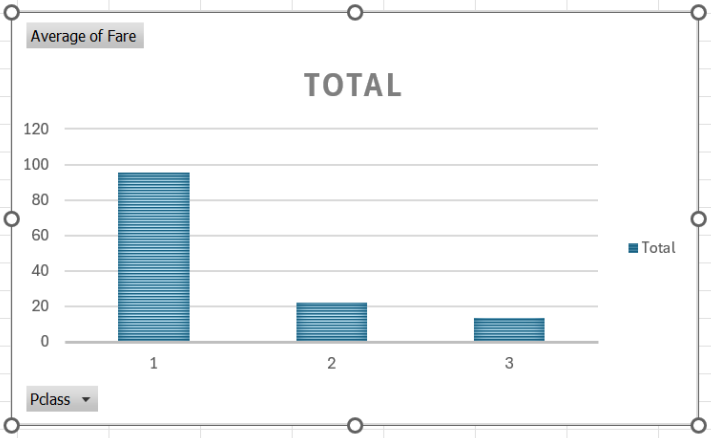
TASK 10: Pivot Chart – Average Fare Visualization

Working

A pivot chart is directly linked to a pivot table. When the pivot table data changes, the chart updates automatically. A column chart visually represents average fare by class.

Implementation

Row Labels	Average of Fare
1	95.60802868
2	22.0557
3	13.69488739
Grand Total	**48.3954076**



Insight

- Clear visual comparison of fares.
- Highlights large fare gap between classes.
- Makes financial differences easy to understand.